PRISMS
Princeton International School of Mathematics and Science

Bridging the Quantum Gap: A Robotic Modeling System for Visualizing Decoherence and Qubit Behaviors

Sruti Nallakukkala
Mentor: Mr. George Heim

Introduction

- Quantum noise causes **compounding errors** with each computation, thus making them **unscalable** and **unreliable**.
- While advances occur in STEM fields, there are not as many opportunities for primary and secondary school students to explore **modern-day advances in STEM**. Furthermore, learning about such advances requires a high pre-requisite level, often causing **significant barriers to entry**.
- Research shows that by being able to **create and apply knowledge** in a way students understand, students are able to **conceptualize** and **retain** information much more effectively.

This research project proposes the use of **a robotic swarm** and **discrete components** to **model qubit behaviors**, to provide a more **engaging introduction** to the **fundamental** concepts of quantum computing.

Results

Part of the challenge of using discrete components is understanding their **capacity** to **model quantum behavior** in the **same** capacity as actual qubits. Through the following graphs, we can compare the capacity of the discrete components using **ideal data** simulated through Qiskit.

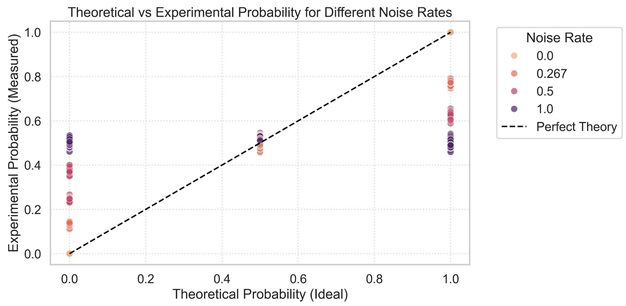


Figure 1
A scatterplot comparing the theoretical and experimental probability obtained for different noise rates [6].

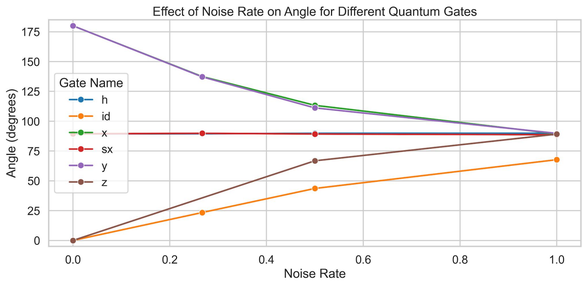


Figure 2
A line graph showing how the angles calculated by the quantum gate are impacted by the noise rate [6].

Conclusion

1. We can use **discrete components** to **model** single-qubit behaviors.
2. Naturally, **measuring** the results of quantum computations introduces quantum noise into the system, noticeable when receiving inaccurate **theoretical** probabilities.
3. As the **noise** in the system increases, the **accuracy** of **quantum computations** decrease significantly, causing these results to be **unreliable**.
4. In the future, knowing that we can use discrete components to **model single-qubit behaviors**, we can use multiple types of discrete components to model **different multi-qubit behaviors,** as well, thus capturing a **wider range** of how qubits interact in relation to one another and to each other.

Materials and Methods

Experiment Methodology:

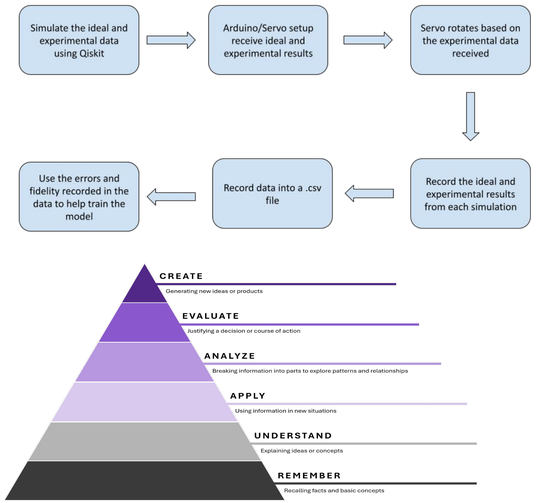
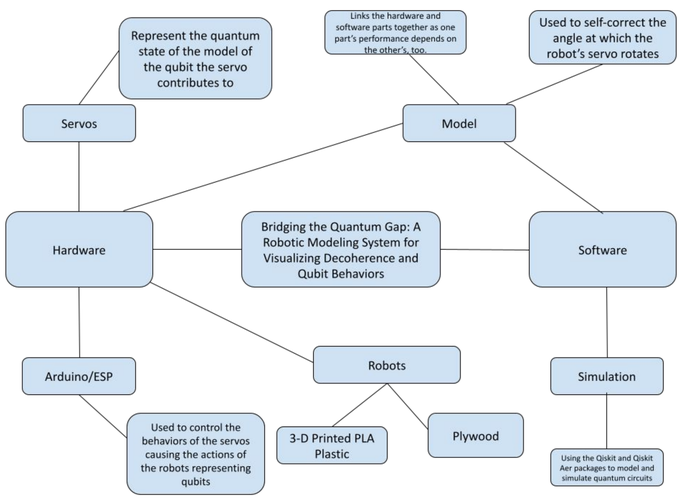


Figure 4
A pyramid depicting Bloom's Taxonomy of Cognitive Skills [3].

Concept Map and Materials:



Discussion

1. The materials used to create qubits have their own **quantum properties**, or have properties that can be considered **analogs** of quantum properties.
2. When using discrete components, because they **do not** have the same abilities as materials with quantum properties, they are **limited** in the **range** of quantum behaviors they could **represent**.
3. Through the results of the experiment, the idea that discrete components can model single-qubit behaviors is **reinforced**.
4. If discrete components can model single-qubit behaviors to a **certain extent of accuracy**, we can use these components to also **simulate** and **model multi-qubit behaviors**, allowing us to use **easily-accessible materials** to **model** interactions **within** and **with qubits** and **the system** they make up.

Future Research

1. Use more discrete components to model **multi-qubit behaviors**, like **entanglement**
2. Design and build **a robotic swarm** to be able to model multi-qubit interactions
3. Train a **model** to be able to **self-correct** the robot to its ideal position in order to **counteract** the noise introduced into the system by **various means**, not only through the simulation itself.

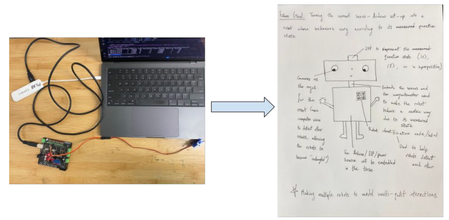


Figure 3
Current experimental setup vs. the end goal of this research project [6].

References

1. ZME Science. (n.d.). What is Bloom's Taxonomy? The pyramid of true learning. *ZME Science*. https://www.zmescience.com/feature-post/resources/school-study/what-is-blooms-taxonomy-the-pyramid-of-true-learning/
2. Hung, S.-H., et al. (2019). Quantitative robustness analysis of quantum programs. Proceedings of the ACM on Programming Languages, 3(POPL), Article 31. https://doi.org/10.1145/3290344
3. "Bloom's Classification of Cognitive Skills," *www.k-state.edu*. https://www.k-state.edu/assessment/toolkit/outcomes/blooms.html
4. Qiskit contributors, "Qiskit: An Open-source Framework for Quantum Computing," 2023. [Online]. Available: https://qiskit.org
5. J. D. Hunter, "Matplotlib: A 2D graphics environment," *Computing in Science & Engineering*, vol. 9, no. 3, pp. 90-95, 2007.
6. S. Nallakukkala, "Bridging the Quantum Gap: A Robotic Modelling System for Visualizing Decoherence and Qubit Behaviors," unpublished research paper, Princeton International School of Mathematics and Science, 2027.

Acknowledgements

A special thanks to the following people:
1. **Mr. Heim** for always being open to my ideas and providing suggestions to help me accomplish the goals for this research project
2. **Gracie Anghelescu** and **Sunny Gao** for motivating me to continue this project, despite the knowledge gap I needed to fill and the unmotivated, directionless days
3. **Echo Ma** and **Sunny Zhang** for providing me feedback on how to better present my research project to other members of the PRISMS community
4. **Mr. Jones** and **Mr. White** for providing me with valuable feedback on my presentation for the last research symposium
5. **Atharva Abhyankar**, **Jerry Gu**, and **Faraz Tamboli** for their interestingly enjoyable antics